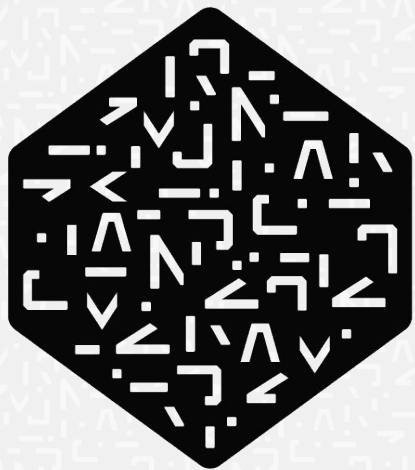




NUMERAI

Michael Oliver
Chief Scientist @ Numerai

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<https://numer.ai/terms>

What is Numerai?

**A market neutral, global equities
hedge fund.**

**The hardest data science tournament
on the planet.**

**Also the highest paying.
\$20M+ paid out.**

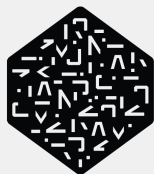
Who is Michael? Chief scientist at
Numerai

Formerly a computational
neuroscientist

What is a neuroscientist doing at a
hedge fund?

Neuroscience data is ...

- Low signal-to-noise
- High dimensional
- Non-linear
- Non-stationary
- Under-sampled



NUMERAI

There are many smart people out there doing amazing things

Numerai unleashes their intelligence on the stock market

id	era	feature1	feature2	target
n2b2e3dd163cb422	era1	0.25	1.00	0.50
n177021a571c94c8	era1	0.75	0.00	0.50
n7830fa4c0cd8466	era1	0.50	0.50	1.00
nc584a184cee941b	era1	0.25	1.00	0.00
nc5ab8667901946a	era1	1.00	0.50	0.75
n84e624e4714a7ca	era1	0.50	0.25	1.00
n6c3f4ad216ac213	era1	0.00	0.70	0.50

Hedge fund quality data
cleaned, obfuscated,
and given away to be
modeled

Train once,
Predict daily

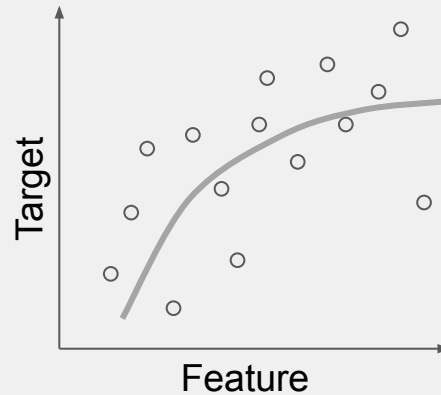
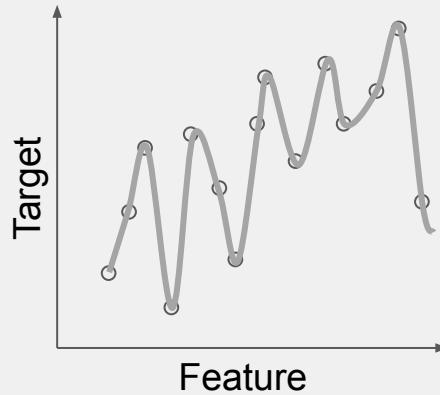
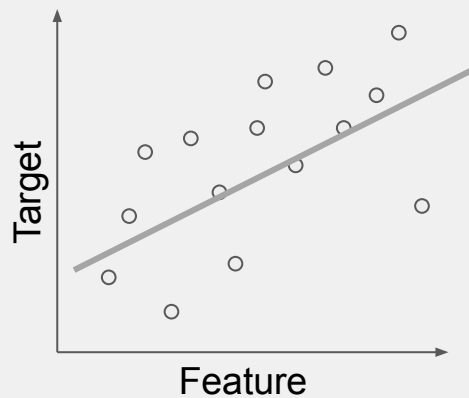
Stake on your
predictions with **NMR**

Earn NMR / Burn NMR

Financial data is way smaller than it seems

- We don't just care if AAPL went up
- We care if AAPL went up and everything else went down
- 1 era = 1 sample
- So we have ~1000 samples, and ~2000 features
- So these dimensions are... terrible for ML

The Art of Machine Learning



Staking

- "Skin in the game" lets us know if a submission is serious + high confidence
- Lets us know how much to weight each user
- Self-correcting feedback mechanism
- Over \$6 million worth of NMR staked

Model Upload for easy automation



```
# Define your prediction pipeline as a function
```

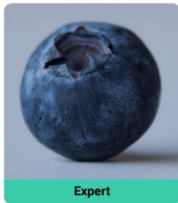
```
def predict(live_features: pd.DataFrame) -> pd.DataFrame:  
    live_predictions = model.predict(live_features[feature_cols])  
    submission = pd.Series(live_predictions, index=live_features.index)  
    return submission.to_frame("prediction")
```



```
# Use the cloudpickle library to serialize your function
```

```
import cloudpickle  
p = cloudpickle.dumps(predict)  
with open("predict.pkl", "wb") as f:  
    f.write(p)
```

We also provide tools for self-hosting models



BLUEBERRY MUFFIN

Joined 4 years ago

A non-random walk down Wall Street.

MMC Rank

6th

of 2,173

CORR Rank

201st

of 2,173

Season Rank

27th

of 395

Stake

10,341 NMR

\$88,000

1Y Return

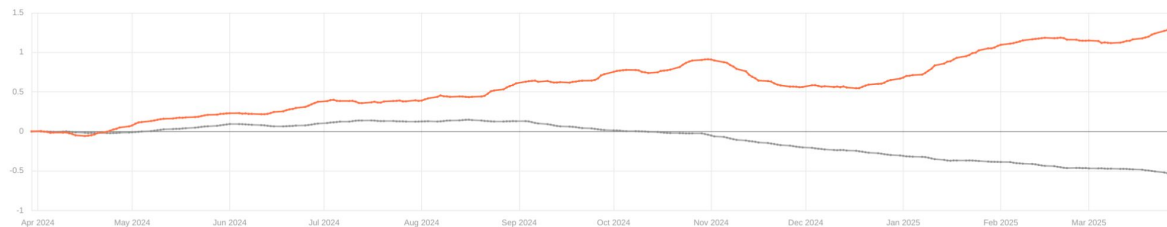
84%

\$33,270.04

MMC

Cumulative stake-weighted-average MMC score of all models over the past 1 year vs Numerai benchmark models

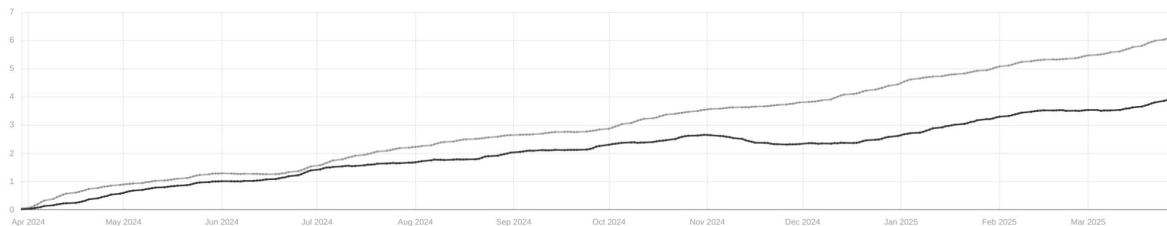
☒ Numerai Benchmark Models



CORR

Cumulative stake-weighted-average CORR score of all models over the past 1 year vs Numerai benchmark models

☒ Numerai Benchmark Models



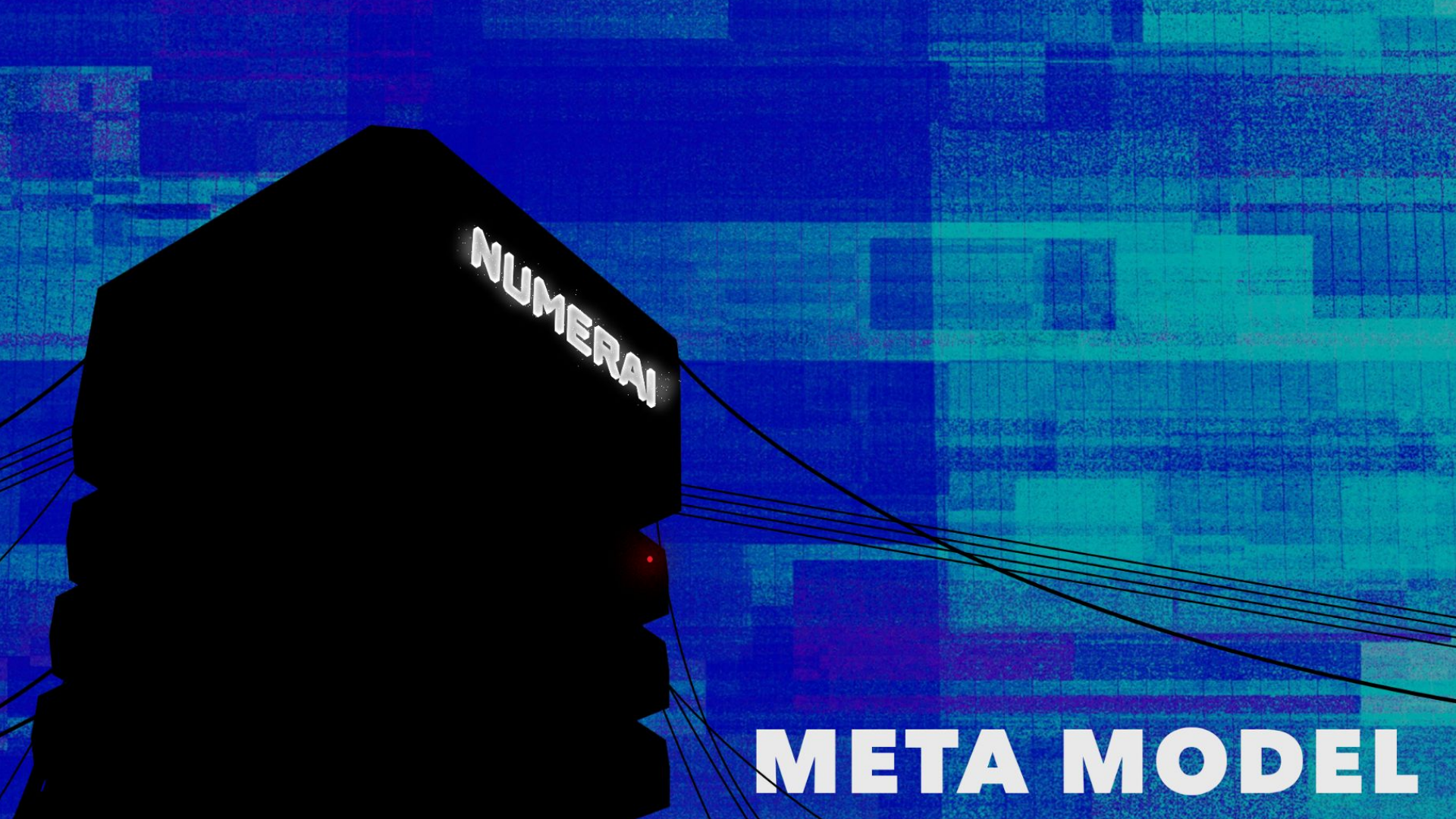
Leaderboard shows
current reputation and
stake values

Scores change daily with
market movements

Payouts happen daily
when 4-week long round
resolves

Predictions from each
user are aggregated,
weighted by stake to
form ...

Data Scientists		Models		Search							
Staked Models		Staked NMR At Risk		Avg 1Y Return							
3,990		585,811 NMR		47.65%							
RANK	NAME	COR360	FNCV3	CORR20V2	▼ MMC	BMC	CORT20	STAKE	1D	3M	1Y
1	 GAMMARAT_076	0.0193	0.0163	0.0163	0.0103	0.0115	0.0101	—	—	—	—
2	 GAMMARAT_083	0.0188	0.0129	0.0152	0.0098	0.0108	0.0113	—	—	—	2.5%
3	 DORA_08	0.0221	0.0184	0.0185	0.0095	0.0121	0.0153	0.026 NMR	0.10%	38.6%	149.2%
4	 FORS002	0.0444	0.0208	0.0230	0.0093	0.0121	0.0207	7.273 NMR	0.96%	47.7%	—
5	 PL11_STANISLAW_LEM	0.0353	0.0228	0.0229	0.0093	0.0125	0.0139	1.137 NMR	2.6%	40.9%	177.4%
6	 GAMMARAT_054	0.0267	0.0120	0.0134	0.0089	0.0100	0.0092	—	—	—	—
7	 GAMMARAT22	0.0172	0.0108	0.0126	0.0089	0.0098	0.0084	—	—	—	22.1%
8	 GAMMARAT36	0.0264	0.0131	0.0146	0.0087	0.0099	0.0103	2 NMR	1.8%	—	—
9	 GAMMARAT_080	0.0157	0.0134	0.0150	0.0085	0.0096	0.0120	—	—	—	76.2%
10	 GAMMARAT33	0.0245	0.0099	0.0142	0.0085	0.0098	0.0110	—	—	—	67.3%
11	 GAMMARAT14	0.0222	0.0144	0.0152	0.0085	0.0097	0.0100	2 NMR	0.10%	—	—
12	 PL0_STEFAN_BANACH	0.0381	0.0196	0.0226	0.0084	0.0119	0.0158	0.086 NMR	3.0%	39.6%	150.5%
13	 MDEL0022	0.0118	-0.0027	0.0094	0.0084	0.0089	0.0012	—	—	—	—
14	 GLAS007	0.0488	0.0213	0.0237	0.0084	0.0113	0.0231	6.026 NMR	2.6%	38.7%	—
15	 GAMMARAT05	0.0143	0.0138	0.0148	0.0083	0.0096	0.0100	—	—	—	—
16	 GAMMARAT_092	0.0168	0.0101	0.0127	0.0082	0.0092	0.0117	2 NMR	1.1%	—	—
17	 GAMMARAT_099	0.0176	0.0116	0.0138	0.0081	0.0090	0.0079	—	—	—	—
18	 GAMMARAT_067	0.0147	0.0124	0.0129	0.0079	0.0089	0.0076	—	—	—	—



NUMERAI

META MODEL

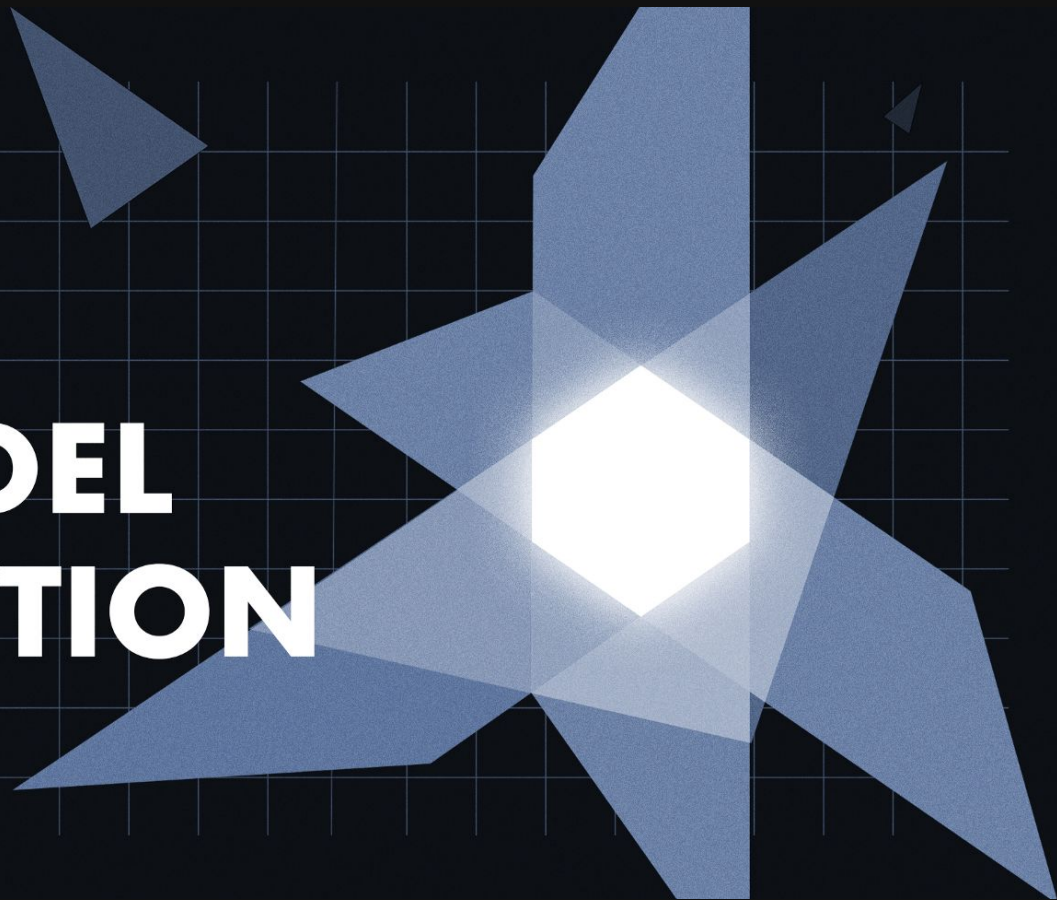
Diversity is King

- "Diversity" in other ML problems isn't very high priority
- For us, it is maybe the highest priority
- Okay performance and high uniqueness is more valuable than high performance but not much different than something we already know
- The market itself is an ensemble of diverse opinions
- So we think the Meta Model is the best way to predict it

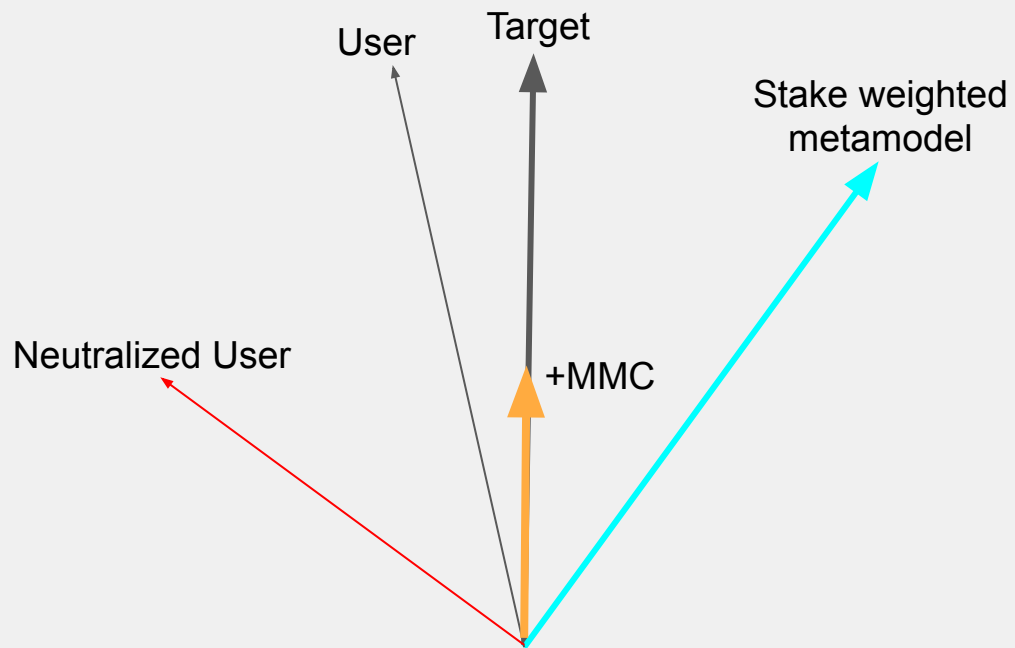


NUMERA I

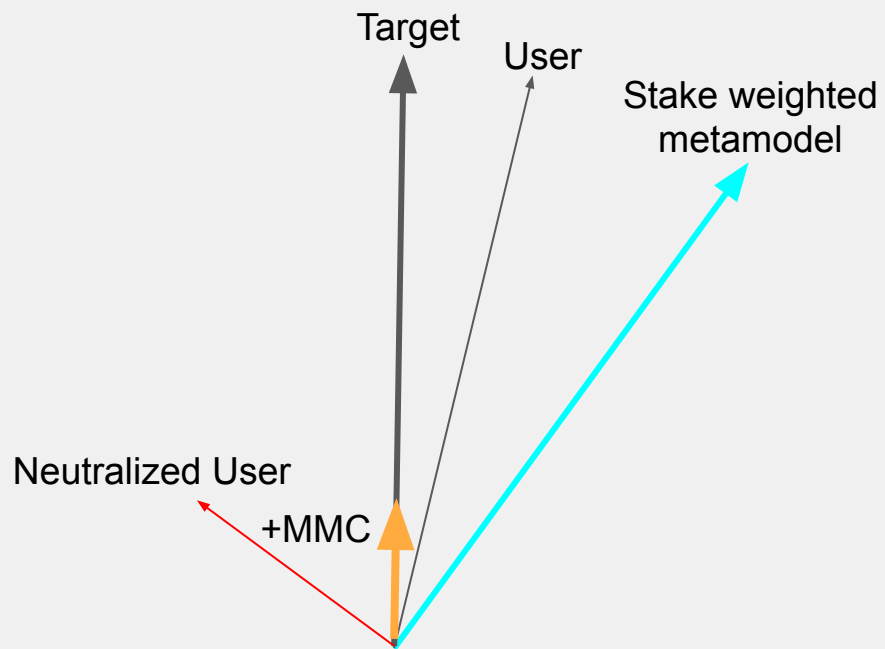
META MODEL CONTRIBUTION



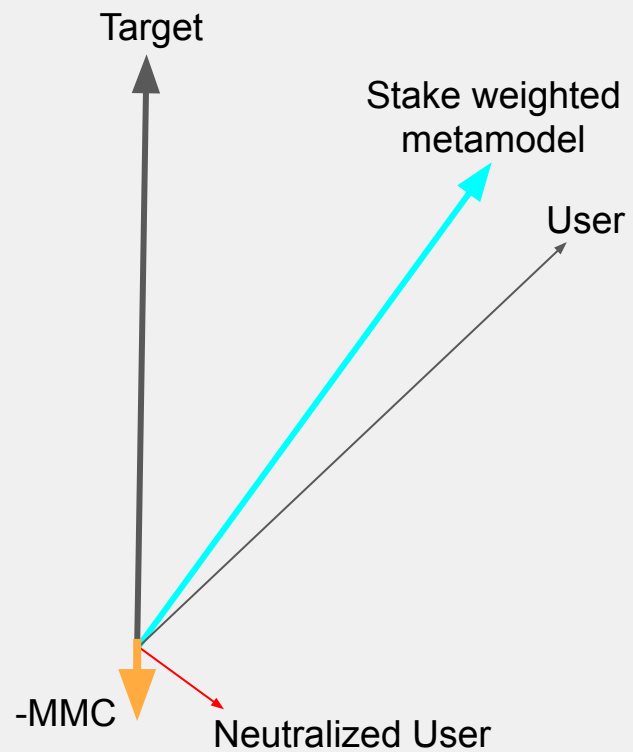
MMC



MMC



MMC



Incentive alignment

Users want to make money – We want to make money

Users want to contribute to the Meta Model prediction of targets

Targets must be amenable to modeling and reflect stock returns

Higher correlation to targets : more profitable portfolios

Lower correlation to targets : less profitable portfolios

Goodhart's law:

"When a measure becomes a target, it ceases to be a good measure"

Why this is hard

Churn – signals that change too quickly can't be traded profitably

Liquidity & market impact – we can't buy full positions in every stock

Volatility & risk – we don't necessarily want full positions in every stock

Why this is hard

These constraints can make signals that have the same correlation to stock returns generate portfolios with very different profitability

Very difficult to incorporate all these constraints into targets that still reflect stock returns and are amenable to modeling

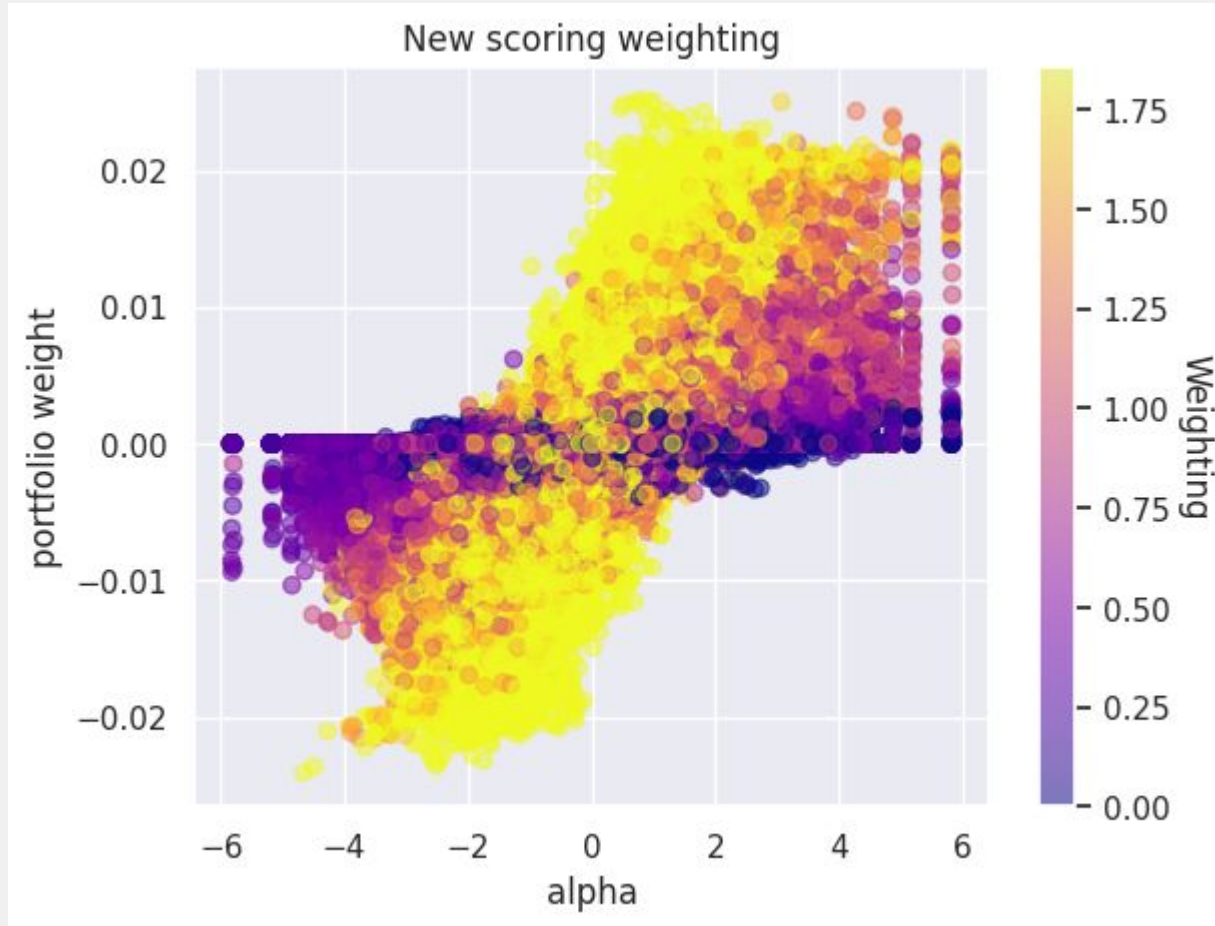
Alignment in Classic vs Signals

In **Classic** we control **targets and features** which helps manage alignment

In **Signals** we only control **targets** which makes alignment much trickier

Wider range of potential submissions =
more potential for **unique information** and **misalignment**

Volatility and Liquidity Weighting



Alpha: A Risk-Aware Scoring System for Signals

Alpha transforms a submission to be **neutral to the risk factors** our portfolio optimizer cares about

The transformation also **weights the predictions based on volatility and liquidity** similar to the optimizer

The Signals scoring system enforces a **turnover limit**

The transformed predictions are essentially a **tradable portfolio**

Alpha uses simple linear algebra and is **user computable**

Meta Portfolio Contribution (MPC)

User portfolio = $\text{Alpha}(\text{User submission})$

Meta Portfolio = $\text{sum}(\text{all stake weights} * \text{all user portfolios})$

Meta Portfolio return = $\text{sum}(\text{Meta Portfolio} * \text{stock returns})$

MPC = $\text{gradient}(\text{Meta Portfolio return}, \text{stake weights})$

Meta Portfolio Contribution (MPC)

Rewards and burns based on MPC directly adjust stakes in the direction of improved Meta Portfolio returns

Users are participating in the online optimization of the Meta Portfolio

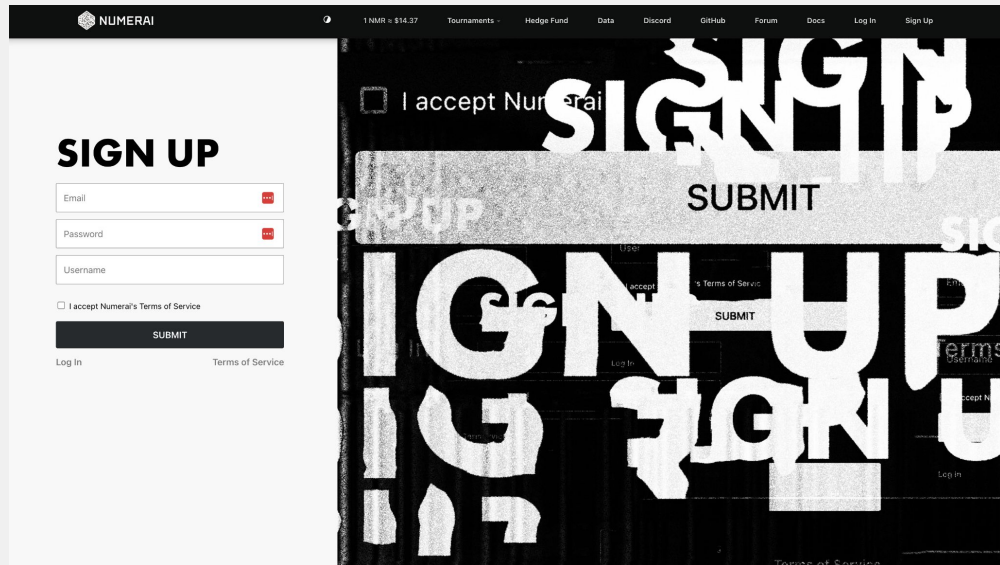
Meta Portfolio is now used to drive trading of our real portfolios

Thanks



Come visit us: [HTTP://NUMER.AI](http://numer.ai)

numer.ai/signup



The image shows a screenshot of the Numerai website's signup page. The page has a dark header with the Numerai logo and navigation links. The main content area is white and features a 'SIGN UP' heading. Below the heading are input fields for Email, Password, and Username. There is a checkbox for 'I accept Numerai's Terms of Service' and a 'SUBMIT' button. The page is overlaid with a large, semi-transparent 'SIGN UP' watermark.

NUMERAI

1 NMR = \$14.37 Tournaments Hedge Fund Data Discord GitHub Forum Docs Log In Sign Up

SIGN UP

☐ I accept Numerai's Terms of Service

SUBMIT

[Log In](#) [Terms of Service](#)



You're almost done

Click this button to activate your Numerai account:

Activate

Or click this link:

[https://numer.ai/confirm/
LPIXM5SQ5PXRDEXATGJGHR75ILONC7](https://numer.ai/confirm/LPIXM5SQ5PXRDEXATGJGHR75ILONC7)

Didn't sign up to Numerai? You can safely ignore this email.

Build your first model

 NUMERAI

1 NMR $\approx$ \$14.36

Tournaments ▾

Hedge Fund

Discord

GitHub

Forum

Docs



Home »

Leaderboard

Data

Models

Submissions

Scores

Staking



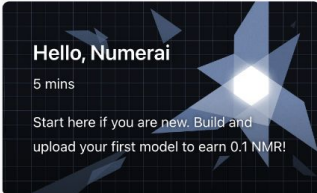
Getting started

Complete these 3 tutorials to learn the basics of the tournament.

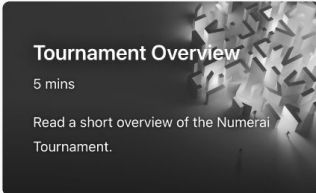
0/3

- [Build your first model](#)
Build your first machine learning model and earn 0.1 NMR
- [Learn about risk](#)
Apply feature neutralization to improve your performance
- [Create an ensemble](#)
Create an ensemble of models trained on different targets

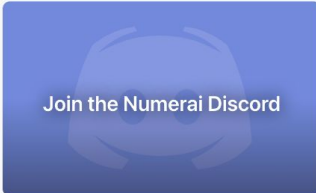
Resources



Hello, Numerai
5 mins
Start here if you are new. Build and upload your first model to earn 0.1 NMR!



Tournament Overview
5 mins
Read a short overview of the Numerai Tournament.



Join the Numerai Discord



View the Numerai F

Open Example Notebook

Hello Numerai

Estimated time: 5 minutes

Build and upload a machine learning model which will automatically submit daily predictions to Numerai.

STEP 1

Run the example notebook

Run the example notebook in Google Colab to download the dataset, train a model, and generate a model file.

Open Notebook

STEP 2

Upload your model

Upload the model file generated in Step 1 to start submitting live predictions to Numerai every day.

Upload Model

STEP 3

Your first submission

Numerai will run your uploaded model to generate your first live submission.